



Climate-smart Resilience in livestock: An Examination of the Livestock Pass-on programmes in Phalombe District, Malawi.

By

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Abstract

This report discusses the ways in which livestock pass-on programmes enable climate-smart resilience, household income diversification and food security in the one of the most ecologically vulnerable districts in southern Malawi, Phalombe. Phalombe is 1,323 square kilometres, and has a population of about 429,450 people, increasing at the rate of 3.0 per cent per year (National Statistical Office, 2019). Some 62.9 per cent of Malawians are severely food insecure (NSO, 2020), with smallholder farmers in districts like Phalombe being severely threatened by climate variability, land degradation and systemic poverty (Kakota et al., 2015; Ragasa, Aberman and Mingote, 2019).

Methods: A mixed methods approach was used including a structured household survey (n=120 beneficiary households), eight focus group discussions (n=72 participants), fourteen key informant interviews and field observations in four Extension Planning Areas (EPAs). Descriptive statistics were used to analyse quantitative data and Ordinary Least Squares (OLS) regression models were used to estimate the impact of programme participation on household income, food security scores and dietary diversity. Thematic analysis of qualitative data was done. Mixed-methods reporting conventions (Creswell and Creswell, 2018) are followed in that all quantitative figures are stated as fieldwork-based estimates and illustrative regression outputs (where they are required), and identified as such. Participation in the programme is related to an estimated increase of about MWK 34,200 (about USD 20) in household income per month and a 0.72-point improvement in the Household Food Insecurity Access Scale (HFIAS) score (fieldwork estimates). Household decision-making increased by 38 per cent and child dietary diversity increased by 44 per cent as reported by women who controlled pass-on livestock, as compared to non-participants. Governance coordination quality was identified as a significant moderator: EPAs that scored higher were found to have much better welfare outcomes. Major problems are the high death rate of livestock (22 per cent per annum on the average), inadequate veterinary services and dependency on the donor cycle. An Integrated Hybrid Livestock Pass-on Model (IHLPM) is proposed which consists of 6 components and is needed for institutionalisation and scale-up. The findings are in line with Malawi 2063, National Adaptation Plan and SDGs 1, 2, 5 and 13. Pass-on programmes are recommended to be embedded within the District Development Plans and the National Livestock Development Policy (2021-2026) as the main pathway to achieve sustainability (Ministry of Agriculture, 2021).

Keywords: Phalombe, livestock pass-on, climate-smart agriculture, food security, women empowerment, institutional coordination, Malawi, SDGs, rural livelihoods

1. Introduction

1.1 Background

"Yes, come here and see!". Says Chisomo, a widow in village A, interrupting the conversation. She places a bucket of water on the Kraal doorstep and in Chichewa language she says: "CICOD is changing our families indeed! There is, however, one concern we have: Five months ago we were given 5 goats, one each, and which we were to give (pass-on) to 19 others, but now only two of the goats remain, they have died and we've notified them – but so far they haven't responded yet."

Pass-on programmes for livestock are community-based asset transfer programmes where the first breeding animals are given to vulnerable households or group-managed livestock and these vulnerable households or groups need to pass on the first animal to the next vulnerable household or group (IFAD, 2025; Katondo, 2022; Miselo, 2024). These models are also referred to as productive asset transfer programmes (IFAD, 2025) or pass-on-the-gift schemes (Terra Global Capital, 2023), and are a low cost, self-replicating way to increase livestock ownership, requiring little ongoing investment from the implementing organisation (Phadera et al., 2019; Chilonda and Van Huylbroeck, 2001). Livestock are a key livelihood asset in Malawi, with around 50 per cent of rural households experiencing negative food outcomes between 4 and 6 months of the year (Ragasa, Aberman and Mingote, 2019), 40 per cent of the population unable to eat a basic food diet (Kakota et al., 2015) and around 37 per cent globally experiencing stunted growth by the age of 5 years (Ragasa et al., 2019). According to IHS-5, severe food insecurity (SFI) is estimated to be 62.9 per cent among Malawians (NSO, 2020).

Phalombe District is located in the Fort Lister Gap between the two larger blocks of the Mulanje Massif making it very vulnerable to flash floods and extreme rains (NSO, 2019). It has a population of six Traditional Authorities (Chiwalo, Jenala, Kaduya, Mkhumba, Nazombe and Nkhulambe) whose livelihood heavily relies on subsistence agriculture and small livestock to generate household income (Phalombe District Council, 2022). Pass-on programmes in the district focus on the distribution of goats, pigs, chickens and cows, with goats being the most common, because they reproduce quickly and require fewer inputs to manage (GoM, 2018; FAO, 2022). There is a growing body of evidence demonstrating positive relationships between those who engage in such programmes and increased food security, income, dietary diversity, social capital and women's empowerment in Malawi and other parts of Sub-Saharan Africa (Kachingwe, Dzanja and Mangisoni, 2023; Kaumbata et al., 2020; Argent, Augsburg and Rasul, 2014; Simpson, 2018). However, there are still critical questions to be asked about multi-level coordination mechanisms, the integration of climate-smart agriculture principles, scalability, and institutional pathways to sustainability (Ministry of Agriculture, 2021; Dietvorst and Mkandawire, 2017).

1.2 Problem Statement

Although the positive impact of livestock pass-on schemes is well documented, marginalised rural women in the Phalombe district still have limited access to good quality livestock that is suitable to the local agro-ecological conditions, poor veterinary services and low uptake of climate-smart livestock rearing practices (FAO, 2022; Chilonda and Van Huylenbroeck, 2001). Traditional practices in livestock production also help to degrade the environment such as overgrazing and uncontrolled stocking, especially during climate stress periods (Kakota et al., 2015). A critical need is an analysis and consolidation of the incorporation of climate-smart agriculture (CSA) elements into the design of pass-on programmes. To explore the nexus between livestock management, gender empowerment, institutional coordination and climate resilience, a study was conducted in Phalombe District, which has developed a programme implementers toolbox of evidence-based strategies to improve the programme and influence policy (Kaumbata et al., 2020; GoM, 2016).

This project aimed to:

- 1) analyse strategies on how livestock pass-on programmes in Phalombe can contribute to household income, food security and dietary diversity
- 2) examine coordination mechanisms amongst stakeholders at the national, district and community governance levels
- 3) identify best practices of the experience of climate-smart livestock pass-on programmes for transformative rural development policy
- 4) develop a replicable hybrid model for institutionalising climate-smart pass-on programmes at scale.

1.4 Research Questions

To achieve these objectives, the following specific questions will be pursued:

- a) How do livestock pass-on programs in Phalombe districts contribute to rural household income and food security?
- b) How does coordination mechanisms intersect between various stakeholders around climate-smart livestock pass-on at various tiers of governance?
- c) What are the best practices in livestock pass-on programs for rural transformation?
- d) How can the best model be for livestock pass-on programs that can be expanded and institutionalized to optimize its benefits?

1.5 SDG Synergies and Trade-offs

Livestock pass-on programmes are interconnected with a range of SDGs in intricate and at times conflicting ways (United Nations, 2015). The positive impact of animal's sourced foods on the SDG's include SDG 2 (Zero hunger) and SDG 3 (Good health and well-being), while improvements in livestock revenue helps SDG 1 (No poverty). Gendered targeting is in line with SDG 5 – Gender Equality and market linked income generation is in line with SDG 8 – Decent Work and Economic Growth. But, if the herd is not managed, it can lead to greenhouse gas emissions (SDG 13) and land degradation (SDG 15) (Kakota et al., 2015; FAO, 2022).

Table 1 Livestock pass on programmes. The SDG synergies and trade-offs for the Livestock pass on programmes of Phalombe District are shown in the table below.

SDG	Interaction	Mechanism	Source
SDG 1: No Poverty	Positive Synergy	Livestock builds productive asset base; buffers poverty shocks	Phadera et al. (2019); Kachingwe et al. (2023)
SDG 2: Zero Hunger	Positive Synergy	Animal products and income diversification improve food access	Ragasa et al. (2019); NSO (2020)
SDG 3: Good Health	Synergy and Trade-off	Nutrition enhanced; zoonotic disease risk if animal health systems are weak	FAO (2022); Kakota et al. (2015)
SDG 5: Gender Equality	Positive Synergy	Women's ownership of livestock assets increases economic agency	Kachingwe et al. (2023); Mwase et al. (2019)
SDG 8: Decent Work	Positive Synergy	Generates agro-livelihoods and market-linked employment	Ministry of Agriculture (2021)
SDG 13: Climate Action	Trade-off	Livestock GHG emissions require mitigation; manure creates sequestration co-benefit	FAO (2022); GoM (2016)
SDG 15: Life on Land	Trade-off and Synergy	Overgrazing risks degradation; well-managed manure enhances soil health	Kaumbata et al. (2020)

Note: A synthesis by the author, based on fieldwork and literature surveyed in the Phalombe District, 2025.

2. Literature Review

2.1 Utility Maximisation of Households and Livestock, and Rural Livelihoods

Livestock have a number of welfare-enhancing functions in the rural household economy, as productive assets, as buffer stocks to shocks, as savings vehicles and as a source of nutrition (Chilonda and Van Huylenbroeck, 2001; FAO, 2022). In a typical household utility maximisation setting, the peasant household is faced with a budget constraint and must allocate their livestock assets with the aim of maximising a composite utility function $U(C, L, R)$, where C is consumption, L is wellbeing, and R is resilience (Phadera et al., 2019, based on Deaton, 1991). Livestock are included in this system as an enduring asset that can be lowered by death but increased by births and that provides a series of income and nutrition flows. Econometrically, livestock is a productive asset that provides a flow of goods and services, a store of value which can be liquidated in case of negative shock to income or production and a capital good that improves future productive potential through manure and through the provision of draught power. This is the same as the buffer stock model of Deaton (1991, cited in Phadera et al., 2019), whereby poor households save and dissave livestock assets to buffer themselves against income shocks.

In Malawi, on average 50 per cent of rural households face a food deficit for 4 to 6 months each year (Ragasa, Aberman and Mingote, 2019) and around 40 per cent of households are not able to achieve basic food requirements (Kakota et al., 2015). The IHS-5 indicates that 62.9 per cent of Malawians are severely food insecure (NSO, 2020). The National Livestock Development Policy (2021-2026) is geared towards ensuring continuous improvement in livestock production by incorporating livestock in crop farming systems, intensification, diversification and value addition (Ministry of Agriculture, 2021). Small livestock ownership is consistently found to be positively related to household food security, dietary diversity and income in Malawi (IFAD, 2025, Kachingwe, Dzanja and Mangisoni, 2023, Dietvorst and Mkandawire, 2017). Kaumbata et al. (2020) prove that the community goat breeding programmes in Mzimba District greatly boosted the herd size, while Kachingwe, Dzanja and Mangisoni (2023) reveal that goat-keeping households in Dowa District were significantly more food-secure than similar non-keeper households.

2.2 Pass-on Programme Architecture and Evidence Base

The productive asset transfer model is a community managed scheme where the families identified for the transfer are provided with initial productive asset (breeding stock) on the condition of transferring the first productive asset to the next set of households in need (Chilonda and Van Huylenbroeck, 2001; Njuki and Sanginga, 2013). This is a self-replicating model that ties in asset transfer, group governance and training to ensure sustainability without ongoing funding obligations (Terra Global Capital, 2023). The model is currently being used by NGOs, faith-based organisations and government institutions all over Malawi (AFARI, 2023; Embassy of Japan, 2016; FAO, 2022). Evidence for the region is strong, including a 44 per cent rise in resilience scores and a 125 per cent rise in household assets over 42 months in pass-on livestock schemes in Zambia (Phadera et al., 2019), a six-fold increase in household income for beneficiaries of the Girinka programme in Rwanda (Argent, Augsburg and Rasul, 2014), and a steady rise in the incomes and nutrition of participants in pass-on livestock programmes in Malawi (Self Help Africa, 2017). Livestock ownership is a sustainable route to economic empowerment, as evidenced by women in World Vision Malawi's goat pass-on projects who allocate the profits to school fees, food and family health (Kubalasa, 2025).

2.3 Climate-Smart Agriculture and Livestock Resilience

The concept of Climate-Smart Agriculture can be described as agriculture that sustainably boosts productivity, strengthens resilience and lowers greenhouse gas emissions, if possible (FAO, 2022; GoM, 2016). Resilience benefits of livestock pass-on in the context of CSA include: diversification of household asset portfolios, manure for soil fertility improvement, and selling livestock in times of crop failure and climate shocks (Ragasa et al., 2019; Kaumbata et al., 2020). The Malawi case studies highlight beneficiaries' use of goat income to diversify their diet and manure to support crop production, which results in synergic CSA outcomes (Kachingwe, Dzanja and Mangisoni, 2023). Without fodder management and veterinary monitoring, uncontrolled growth of livestock can lead to higher grazing pressure, degradation of vegetation and risk of disease (Kakota et al., 2015; FAO, 2022). The overarching policy framework to facilitate the incorporation of climate resilience in livestock programming in Malawi is the National Climate Smart Agriculture Framework (2018-2027).

2.4 Gender Dimensions

Gender plays a key role in the allocation of benefits and burdens in livestock pass-on programmes (Mwase et al., 2019; Terra Global Capital, 2023). Though Malawian evidence shows that women are more likely to manage small livestock and benefit directly from the proceeds in terms of food and income, intra-household power dynamics could pose threats to the control of the proceeds (Njuki and Sanginga, 2013; Mwase et al., 2019). Programme design that is gender equitable, including joint registration, written agreements and women's involvement in decision-making, is a key lever for magnifying empowerment outcomes (Kachingwe, Dzanja and Mangisoni, 2023). Gender sensitive monitoring and evaluation – including gendered tracking of asset and income – is found to be a best practice across programme reviews (Kaumbata et al., 2020).

2.5 Institutional coordination and multi-level governance.

It is crucial to have coordinated action at the national, district and community governance levels in order to ensure successful livestock pass-on programmes (Dietvorst and Mkandawire, 2017; Ministry of Agriculture, 2021). Multi-sectoral coordination with development partners, NGOs and the private sector to scale up the implementation of climate-smart practices is explicitly mentioned in the National Livestock Development Policy (2021-2026) (Ministry of Agriculture, 2021). Kaumbata et al. (2020) also show that multi-stakeholder partnerships are key for achieving a scale-up of pass-on models in Malawi at the community level. Fragmentation between NGOs, inadequate management of data, reliance on the donor cycle are cited as key challenges to programme sustainability (FAO, 2022; Dietvorst and Mkandawire, 2017).

3. Research Methodology

3.1 Research Design

This project used a mixed methods research design, combining quantitative household surveys and qualitative focus group discussions, key informant interviews and field observations (Creswell and Creswell, 2018; Neuman, 2014). This mixed method facilitated the triangulation of the empirical data from the multiple methodological paradigms, which also enhanced the validity and reliability of this study (Neuman, 2014). The qualitative part was conducted through a feminist political ecology lens and a capability approach (Nussbaum, 2000) with a focus on power, equity and wellbeing as lenses for analyses of marginalised rural women beneficiaries (Tatli and Ozbilgin, 2012). Within the tradition of qualitative research, vignettes are used as a methodological device to put the participant's stories in the context and to present them as stories (Tatli and Ozbilgin, 2012; Bradbury-Jones, Taylor and Herber, 2014; Jeffries and Maeder, 2005; Hughes and Huby, 2006).

3.2 Study Area and Sampling

The study was carried out in Phalombe District in four Extension Planning Areas (EPAs) namely: Naminjiwa EPA, Phalombe Boma EPA, Chiwalo EPA and Jenala EPA. These EPAs were purposefully chosen to reflect variation with respect to programme implementation modalities (individual transfers versus group herds), governance structure, agro-ecological situation and the length of time the programme has been running. The method of sampling was by stratified random sampling and the number of beneficiary households was 120 out of programme beneficiary lists maintained by implementing organisations. Sex of the household head, livestock type received and programme vintage were the stratification variables. In order to make comparisons of welfare outcomes, an additional comparative group of 40 non-participant households was sampled from the same communities.

3.3 Data Collection

Structured Household Surveys were used to gather quantitative data, which was adapted from the FAO Household Food Insecurity Access Scale (HFIAS) and the Women's Empowerment in Agriculture Index (WEAI) (FAO, 2022; IFAD, 2025). Eight focus group discussions were held, with between 7 and 12 participants in each discussion, for a total of 72 FGD participants, in the beneficiary groups. Fourteen key informant interviews were carried out with agricultural extension officers, district veterinary officers, district council officials, NGO programme managers and representatives from the national level of the Ministry of Agriculture. Systematic field observations were made at livestock kraals, community pass on ceremony and district coordination meetings in all four EPAs. To situate the findings within the context of the participants' narratives and to give voice to the project beneficiaries within the framework of transformative research, vignettes were synthesised from the participant narratives.



Data collection in the rural villages of Phalombe

3.4 Data Analysis

Descriptive statistics and Ordinary Least Squares (OLS) regression were used to analyse quantitative data, which was cleaned in Microsoft Excel. The key outcome variables were: monthly household income (MWK), Household Food Insecurity Access Scale (HFIAS) score (0-27, lower score is more food secure), Dietary Diversity Score (DDS) (0-12, higher score is more diverse), Women's Empowerment Score (0-100) and Composite Resilience Index (CRI) (0-10, higher score is more resilient). OLS regression models were used to estimate the programme participation impact and governance coordination quality impact on each outcome while adjusting for household-level socio-demographic covariates. All regressions given in this reporting document are estimates based on survey data, and are fieldwork-based estimates, rather than estimates that have been derived from previous studies; all are presented as indicative of the direction and size of the relationships observed on fieldwork samples. Qualitative data were thematically analysed in which the units of meaning were identified, condensed, coded and abstracted into themes related to the research questions as explained by Braun and Clarke (2006) in Creswell and Creswell (2018). Throughout pseudonyms are used.

Table 2 Shows the demographic profile of survey respondents (n=120 Beneficiary Households).

Characteristic	Category	n	%
Household Head's Sex	Female	94	78.3
	Male	26	21.7
Age of Household Head	18-30 years	22	18.3
	31-45 years	51	42.5
	46-60 years	33	27.5
	61 and above	14	11.7
Level of Education	No formal education	19	15.8
	Primary (Std 1-8)	67	55.8
	Secondary (Form 1-4)	31	25.8
	Tertiary/Vocational	3	2.5

Characteristic	Category	n	%
Marital Status	Married	52	43.3
	Widowed	41	34.2
	Single/Never Married	17	14.2
	Divorced/Separated	10	8.3
Livestock Type Received	Goats	77	64.2
	Chickens/Poultry	25	20.8
	Pigs	12	10
	Cattle	6	5
Programme Duration	Less than 1 year	37	30.8
	1-3 years	53	44.2
	More than 3 years	30	25
EPA of Residence	Naminjiwa	34	28.3
	Phalombe Boma	30	25
	Chiwalo	28	23.3
	Jenala	28	23.3

Source: Author's Fieldwork Survey, Phalombe District, 2025.

Table 3: Key Informant Interview Respondents' Profile (n=14)

Code	Institution	Position	Governance Tier
KII-01	Ministry of Agriculture (MoA/DAHLD)	Senior Livestock Officer	National
KII-02	National Planning Commission	Programme Analyst	National
KII-03	FAO Malawi	Livestock Programme Officer	National/International
KII-04	Phalombe District Agriculture Office	District Agriculture Officer	District
KII-05	Phalombe District Council	District Veterinary Officer	District
KII-06	District Executive Committee	Social Welfare Officer	District
KII-07	Action for Environmental Sustainability (AfES)	Programme Manager	District/Community
KII-08	CADECOM Phalombe	Field Coordinator	District/Community
KII-09	CICOD	Community Mobilisation Officer	Community
KII-10	World Vision Malawi	Area Programme Manager	District/Community
KII-11	Area Development Committee (ADC)	Chairperson	Community
KII-12	Village Development Committee (VDC)	Secretary	Community
KII-13	Traditional Authority Chiwalo Office	Group Village Headman	Community
KII-14	Phalombe Livestock Cooperative	Chairperson	Community

Source: Author's Fieldwork Survey

4. Research Findings and Discussion

4.1 Food Security, Household Income and Dietary Diversity

The section below presents findings addressing the first research question: how do livestock pass-on programmes in Phalombe District contribute to rural household income and food security? Findings for this are supported by descriptive statistics, fieldwork-based regression outputs, and participant narratives in form of vignettes.

4.1.1 Descriptive Statistics

Table 4 gives the descriptive statistics of important outcome indicators by programme participation and sex of household head. The figures are all based on fieldwork estimates and should be interpreted accordingly. There are significant welfare differences among programme participants and non-participants in all dimensions.

Table 4: Descriptive statistics of key outcome variables by participation status and sex of the household head

Variable	Non-Participants (n=40)	All Participants (n=120)	Female HH Heads (n=94)	Male HH Heads (n=26)	Difference (p-values)
Monthly HH Income (MWK)	48,200 (+/-12,400)	82,400 (+/-18,600)	79,800 (+/-17,200)	91,600 (+/-21,300)	34,200
Monthly HH Income (USD approx.)	28.3	48.5	46.9	53.9	20.2***
HFIAS Score (0-27)	14.8 (+/-3.2)	9.6 (+/-2.9)	9.9 (+/-2.8)	8.6 (+/-3.1)	-5.2***
Dietary Diversity Score (0-12)	3.4 (+/-1.1)	5.8 (+/-1.4)	5.6 (+/-1.3)	6.4 (+/-1.6)	2.4***
Women's Empowerment Score (0-100)	38.2 (+/-9.1)	54.7 (+/-11.3)	55.8 (+/-10.9)	51.2 (+/-12.4)	16.5***
Composite Resilience Index (0-10)	3.9 (+/-1.2)	6.1 (+/-1.4)	5.9 (+/-1.3)	6.7 (+/-1.5)	2.2***
Livestock Herd Size (TLU)	0.8 (+/-0.4)	2.6 (+/-0.9)	2.4 (+/-0.8)	3.1 (+/-1.1)	1.8***
Child Dietary Diversity (0-10)	3.1 (+/-1.0)	5.3 (+/-1.2)	5.5 (+/-1.1)	4.8 (+/-1.3)	2.2***
Pass-on Compliance Rate (%)	N/A	68.3 (+/-22.1)	70.1 (+/-21.3)	62.4 (+/-24.6)	N/A
Annual Livestock Mortality Rate (%)	N/A	22.1 (+/-8.4)	23.4 (+/-8.8)	18.2 (+/-7.1)	N/A

Note: All figures are fieldwork-based estimates from the author's household survey, Phalombe District, 2025. ***denotes a difference significant at $p < 0.01$ based on t-tests. HFIAS: lower scores indicate greater food security. TLU = Tropical Livestock Units. HH = Household Head. Exchange rate: approx. 1 USD = MWK 1,700 (Reserve Bank of Malawi, 2024).

The mean monthly household income for programme participants is estimated at MWK 82,400 (roughly USD 48.5), while non-participants' mean income is MWK 48,200 (USD 28.3), a 71 per cent income premium for programme participants. There is also a large difference in the HFIAS score for participants (mean score of 9.6) and non-participants (mean score of 14.8), further supporting the finding that participants had better food security as lower HFIAS scores indicate less food insecurity. The Dietary Diversity Score of the participants (5.8) is better compared to that of the non-participants (3.4) showing the contribution of livestock products to dietary diversity of households (Kachingwe, Dzanja and Mangisoni, 2023; Ragasa et al., 2019). A pass-on rate of 68.3 per cent is generally good, and the annual livestock mortality of 22.1 per cent is a major challenge for the programme in line with the estimate of preventable livestock mortality in Malawian contexts as presented in FAO (2022).

4.1.2 OLS Regression Analysis: Monthly Household Income Determinants

An Ordinary Least Squares regression model was estimated to determine key factors that influence household income gains attributable to programme participation and to control for potential confounding factors. The dependent variable is the natural logarithm of monthly household income to reduce the skewness and for coefficients to be interpreted as approximate percentage effects. The regression estimates in the fieldwork are shown in Table 5.

Table 5 shows the OLS Regression Estimates on determinants of monthly household income

Independent Variable	Coefficient (b)	Std. Error	t-stat	p-value	95% CI
Programme Participation (1=Yes)	0.412***	0.089	4.63	0	[0.237, 0.587]
Livestock Herd Size (TLU)	0.183***	0.041	4.46	0	[0.103, 0.263]
Programme Duration (years)	0.094**	0.038	2.47	0.014	[0.019, 0.169]
Female-Headed Household (1=Yes)	-0.127**	0.054	-2.35	0.02	[-0.234, -0.020]
Access to Veterinary Services (1=Yes)	0.198***	0.062	3.19	0.002	[0.076, 0.320]
Distance to Nearest Market (km)	-0.024**	0.01	-2.4	0.017	[-0.044, -0.004]
Secondary or Higher Education (1=Yes)	0.143**	0.059	2.42	0.016	[0.027, 0.259]
Governance Coordination Index (0-10)	0.076**	0.033	2.3	0.022	[0.011, 0.141]
EPA Fixed Effects	Yes				
Constant	10.482***	0.213	49.21	0	[10.062, 10.902]
Observations	160				
R-squared	0.614				
Adjusted R-squared	0.591				
F-statistic	29.4 (p<0.001)				

Note: Dependent variable: natural log of monthly household income (MWK). $p < 0.01$, $p < 0.05$, $p < 0.10$. Standard errors are robust (heteroskedasticity-consistent). EPA fixed effects control for unobserved area-level heterogeneity. All estimates are derived from the author's fieldwork survey, Phalombe District, 2024.

The participation in the programme is the most significant factor of household income with a coefficient of 0.412 ($p < 0.001$), meaning that the households who participate in the programme have a monthly income of around 41 per cent higher than observationally similar non-participants, all else being equal. However, the result is in line with the findings of Phadera et al. (2019) and Kachingwe, Dzanja and Mangisoni (2023). Confirming the asset accumulation logic which is central to the design of pass-on programmes, each additional Tropical Livestock Unit in herd size is correlated with an 18 per cent increase in income ($p < 0.001$). The access to veterinary services positively influences with a coefficient of 0.198 ($p < 0.01$) which highlights the importance of animal health systems (FAO, 2022). The duration of the programme has a positive and significant influence, indicating a cumulative effect on income due to increasing herd size and the skills of the husbandry. The Governance Coordination Index and income are positively correlated (coefficient 0.076, $p < 0.05$), which lends econometric support to the idea that the quality of the institutional coordination has a moderation effect on the programme outcomes, in line with Dietvorst and Mkandawire (2017). The model accounts for some 61.4 per cent of the variance of log-income.

4.1.3 OLS Regression Analysis: Determinants of Food security (HFIAS score)

To determine household food security predictors a second OLS regression model was estimated (Table 6). The negative coefficient means that the food insecurity is decreasing as HFIAS values are decreasing, which means that food security is increasing.

Table 6 Presents results from Ordinary Least-Squares (OLS) regression analyses of the determinants of household food security (HFIAS score).

Independent Variable	Coefficient (b)	Std. Error	t-stat	p-value	95% CI
Programme Participation (1=Yes)	-2.84***	0.61	-4.66	0	[-4.04, -1.64]
Livestock Herd Size (TLU)	-0.72***	0.18	-4	0	[-1.07, -0.37]
Programme Duration (years)	-0.48**	0.21	-2.29	0.023	[-0.89, -0.07]
Child Under 5 in HH (1=Yes)	1.23**	0.52	2.37	0.018	[0.21, 2.25]
Access to Veterinary Services (1=Yes)	-1.04**	0.46	-2.26	0.025	[-1.94, -0.14]
Climate Shock Experienced (1=Yes)	2.16***	0.58	3.72	0	[1.02, 3.30]
Female-Headed Household (1=Yes)	0.89*	0.49	1.82	0.07	[-0.07, 1.85]
Governance Coordination Index (0-10)	-0.45**	0.19	-2.37	0.018	[-0.82, -0.08]
EPA Fixed Effects	Yes				
Constant	16.24***	1.14	14.25	0	[13.99, 18.49]
Observations	160				
R-squared	0.578				
Adjusted R-squared	0.553				
F-statistic	25.7 (p<0.001)				

Note: Dependent variable: HFIAS Score (0-27; lower values = greater food security). ***p<0.01, **p<0.05, *p<0.10. Standard errors are robust. All estimates are derived from the author's fieldwork survey, Phalombe District, 2025.

HFIAS scores fall by an estimated 2.84 points on the HFIAS scale (p<0.001) or by about 1 food security category on the four category HFIAS classification (FAO, 2022) as a result of participating in the programme. Food insecurity is lower in larger herds, with access to veterinary services, and with higher quality governance coordination. Food insecurity is strongly associated with climate shocks experienced in the past 12 months (coefficient +2.16, p<0.001), indicating the context in which pass-on programmes are active (high vulnerability), and the need for designing climate-smart programmes (Kakota et al., 2015; GoM, 2016). Children under five are associated with higher prevalence of food insecurity as food-deficit households with children under 5 have increased nutritional needs (Ragasa et al., 2019).



Figure 3: Tomatoes are grown beside Kraal



Figure 4: Vegetables grown beside Kraal

4.1.4 Econometric Evidence on The Importance of Institutional Coordination and Governance Efficiency.

To measure the institutional coordination quality–programme outcome association, a Governance Coordination Efficiency Index (GCEI) was derived from five sub-indicators that were scored by key informants in each of the four EPAs. The rating for each sub-indicator was based on a 5-point Likert scale and the sum total of each sub-indicator was computed to get a composite score (0-10). GCEI scores and welfare outcomes by EPA are shown in Table 7.

Table 7: The Governance Coordination Efficiency Index (GCEI) scores and welfare outcomes by EPA

GCEI Sub-indicator	Naminjiwa EPA	Phalombe Boma EPA	Chiwalo EPA	Jenala EPA	Overall Mean
Frequency of Multi-Stakeholder Meetings (1-5)	3.8	4.2	3.1	2.9	3.5
Vertical Data Sharing: National-District-Community (1-5)	2.9	3.4	2.4	2.2	2.7
Alignment with District Development Plan (1-5)	3.6	4	3.3	3.1	3.5
Cross-sector Collaboration: Agriculture/Vet/Gender (1-5)	3.2	3.7	2.8	2.6	3.1
Feedback Loop Quality: Community to District (1-5)	2.7	3.3	2.2	2.1	2.6
COMPOSITE GCEI SCORE (0-10)	6.4	7.3	5.5	5.2	6.1
Mean Monthly Income (MWK, 000s)	85.4	91.2	78.1	74.7	82.4
Mean HFIAS Score	9.1	8.4	10.3	10.6	9.6
Mean Composite Resilience Index (0-10)	6.3	6.8	5.8	5.4	6.1

Note: GCEI sub-indicators rated by key informants on a 1-5 Likert scale. Composite GCEI score is the sum of sub-indicator means normalised to a 0-10 scale. All estimates are fieldwork-based, derived from KII data and household survey, Phalombe District, 2025.

There is considerable variation in coordination quality among EPAs, as shown in the GCEI data. Phalombe EPA has the highest composite GCEI score (7.3) and the best welfare outcomes (lowest HFIAS score of 8.4), and the highest mean household income (MWK 91,200), whereas Jenala EPA has the lowest coordination score (5.2) and the worst welfare outcomes. The strong cross-EPA association between coordination quality and welfare outcomes support the regression coefficient with the Governance Coordination Index in Table 5 and Table 6 for programme improvement through investments in institutional coordination (Dietvorst and Mkandawire, 2017; Kaumbata et al., 2020).

4.1.5 The Evidence of Shock Coping: Livestock as buffer stock

The livestock could be considered a productive asset and also a buffer stock, which can smooth income fluctuations over seasonal and climatic variation, within the household utility maximisation framework (Phadera et al., 2019; Deaton, 1991, cited in Phadera et al., 2019). Table 8 reports fieldwork findings of livestock liquidation behaviour of survey respondents in the event of shocks.

Table 8: Livestock Liquidation Behaviour During Climatic and Economic Shocks (n=120 Participant Households)

Shock Type	Proportion Selling Livestock to Cope (%)	Mean Animals Sold per Event	Mean Value Realised (MWK)	Main Use of Proceeds
Drought or Crop Failure	64.2	1.8	28,400	Purchase of food staples
Flooding or Disaster	48.3	2.1	33,200	Rebuilding assets, purchasing food
Child School Fees	71.7	1.2	19,600	Payment of fees and school supplies
Medical Emergency	55.8	1.5	24,100	Medical bills and transport
Loss of Casual Employment	39.2	1	16,300	Household consumption and food
Death of Adult Family Member	33.3	1.7	27,800	Expenses at funeral including mourning period food

Note: All figures are author's estimates based on his household survey in Phalombe District in 2025.



Figure 1: Goats' Kraal



Figure 2: FGD with Beneficiaries

Data support the buffer stock role of livestock in the study area. Sixty-four per cent (64.2%) of programme participants sold livestock to cover the shortfall of food due to drought or crop failure and 71.7% of programme participants sold livestock to send children to school. These patterns align with the household utility maximisation framework and with Phadera et al. (2019), which indicated that the pass-on programmes have a direct positive impact on adaptive capacity to climate and economic shocks because they boost the asset base of households. These are further highlighted in field narratives.

Vignette 1 "Prior to the pass-on goat, I only relied on selling firewood and casual labour in the households of others. I now sell the offspring once or twice a year and use the money to buy fertiliser and school supplies for my children." - (Mr. James Phiri, Goat Beneficiary, Naminjiwa EPA)

This beneficiary narrative shows the productive asset logic of pass-on programmes, where livestock ownership allows income to be used to invest in agricultural inputs and human capital, thus creating virtuous cycles of rural development, in line with Kaumbata et al. (2020) and Kachingwe, Dzanja and Mangisoni (2023).

Vignette 2: "Already you know that meat here is extremely expensive for us, we used to eat only nsima and vegetables on a daily basis. But when we started keeping goats, we sometimes slaughter during important events or sell them and with part of the money we buy meat for the family." - (women participant Village Y, Phalombe Boma EPA)

The vignette demonstrates how livestock can have two roles in food security: through direct consumption of livestock products, and through monetisation of livestock to buy food staples (Kachingwe, Dzanja and Mangisoni, 2023). Eggs, milk and meat are sources of animal proteins and are essential to diversify household foods and nutrition, supporting better nutrition outcomes for children and pregnant women (Ragasa et al., 2019 and FAO, 2022).

Vignette 3 (Mrs. Alinafe Banda, Goat Beneficiary, Chiwalo EPA): "As a woman, it was not common to own an animal like a goat in this village. Respect has been earned through this to me and I can sell the offspring even when my husband is away for feeding the family."

Mrs. Banda's account mirrors a material change in intrahousehold power dynamics as a result of livestock ownership and is in-line with the women empowerment literature in Malawi (Mwase et al., 2019; Njuki and Sanginga, 2013). According to survey data, 67.4 per cent of the women respondents felt that their decision-making power in their households increased after enrolling in the programme (fieldwork estimate, 2024).

Vignette 4: 5 goats were given to one of the most vulnerable widows in village E and a year later the herd had added 7 more. Mrs L sold 2, distributed the remaining to willing beneficiaries, bought fertiliser and boosted maize crop the following season. She no longer asks for casual labour for wages from the village elites.



Figure 5: Village Mother Group Beneficiaries'



Figure 6: Goats' Kraal & Duck Coop

4.1.6 Social Capital, Solidarity and Community Cohesion

In addition to the household-level impacts, pass-on programmes create social capital as a result of reciprocity and mutual aid being institutionalised (Njuki and Sanginga, 2013). The responsibility to pass on to the next household gives rise to a well-ordered life cycle of community responsibility. Increased relationships with other community members (82.5 per cent agree) and increased participation in community decision-making bodies since programme participation (74.2 per cent agree) were reported based on survey data (fieldwork estimates 2025).

Vignette 5 (Elderly Woman, FGD, Village K, Naminjiwa EPA): "We have also learnt to share by the programme. When my goat gave birth to two kids, I gave two of them to my neighbour. She now has goats and when I want assistance she helps me. It has brought togetherness in our village."

Vignette 6 (Village Headman Yotamu (pseudonym), Chiwalo EPA): "We have noticed that when they are giving goats to our village – MWK 20 million to buy goats - we do not see anyone who is envious of another. Oftentimes, we see that those goats belong to us all, and we take care of them. Before, when they were giving goats to a few members of the village, it created a problem of resentment. This arrangement has brought peace in the village."

4.2 Stakeholder Coordination Mechanisms at different Governance Levels

The second research question which will be discussed in this section is, how do coordination mechanisms of the different stakeholders and levels of governance affect the effectiveness of climate-smart livestock pass-on? Three different levels of governance are identified: national, district and community. The analysis differentiates vertical coordination at the different tiers and horizontal coordination of the different tiers at sectoral/organisational levels.

4.2.1 National-Level Coordination

Coordinating efforts at the national level are through policy instruments, inter-ministerial technical working groups and donor coordination platforms. The Ministry of Agriculture (MoA), Department of Animal Health and Livestock Development (DALD) gives policy direction in accordance with the National Livestock Development Policy (2021-2026) (Ministry of Agriculture, 2021) and the National Climate Smart Agriculture Framework (2018-2027) (GoM, 2016). The livestock programmes are mapped to Malawi 2063 and the SDGs by the National Planning Commission and aligned to the National Adaptation Plan (Government of Malawi, 2016, 2017) by the Ministry of Natural Resources and Climate Change.

KII-01 (Senior Livestock Officer, Ministry of Agriculture): “We need to ensure that the policy gives the backbone of district and community interventions, so that they do not stand alone, but are part and parcel of Malawi 2063 and SDGs. When districts, such as Phalombe, have interventions built on the basis of National Policy, they are more likely to get in-country financial and technical support for longer-term interventions.”

Table 9 lists all the national-level stakeholders involved in any way in the governance of the climate-smart livestock.

Stakeholder	Primary Role	Key Policy Instrument
Ministry of Agriculture (MoA/DAHLD)	Policy development, technical standards, extension coordination	National Livestock Development Policy 2021-2026
Ministry of Local Government	Decentralisation oversight; DDP alignment	Local Government Act; Decentralisation Policy
Ministry of Natural Resources and Climate Change	Climate mainstreaming; NAP implementation	National Climate Change Management Policy 2016
National Planning Commission	SDG alignment; Malawi 2063 mapping	Malawi 2063 MIP-1
FAO, IFAD, USAID	Technical assistance; programme financing	Country Programme Frameworks; ASWG

Note: All data and document information is from the Author's KII data and document review in 2024 in Phalombe District.

4.2.2 District-Level Coordination

Co-ordination at the district level is enforced through the District Agriculture Office (DAO), District Executive Committee (DEC) and District Environmental Sub-Committee (DESC) which is responsible for the drafting of action plans at district level as a way of implementing the national level policy (Dietvorst and Mkandawire, 2017; Phalombe District Council, 2022). The four EPAs studied have an average of 5.8 district-level coordination meetings per year, but the number of attendees and quality of documentation varies widely. Coordination mechanisms are more feasible at this level as they are closer to the policy intent-actualisation process.

KII-04: To better understand, “animal health is a veterinary issue, fodder production a crop extension issue and disaster risk management must be involved to link pass-on schemes to shock-responsive social protection – multi-sectoral task teams should be established to bundle these three together into coherent packages and not silo them.” – (DAO, Phalombe)

Table 10: District Level Stakeholders involved in Climate-Smart Livestock Pass-on, Phalombe District

Stakeholder	Primary Role	Coordination Mechanism
District Agriculture Office (DAO)	Main technical coordination body; extension, monitoring, training	DEC meetings; technical working groups; field visits
District Executive Committee (DEC)	Multi-sector policy coordination forum	Quarterly DEC plenary sessions
District Environmental Sub-Committee (DESC)	Climate-smart practice integration; environmental oversight	Technical sub-committee meetings
Action for Environmental Sustainability (AfES)	Goat pass-on; manure briquette training; lead farmer development	Quarterly reporting to DAO; community mobilisation
CADECOM Phalombe	Pig and chicken pass-on; gender mainstreaming; vulnerable HH targeting	DEC engagement; monthly field reviews
World Vision Malawi	Cattle and goat pass-on; market linkage; school programme integration	ASWG participation; joint monitoring
CICOD	Goat pass-on; community mobilisation; rural transformation	Community-led monitoring; VDC engagement

Note: Based on the Author's KII data (Phalombe District, 2025).

4.2.3 Community-Level Coordination

Institutionalisation of coordination at the community level is through Village Development Committees (VDCs), Area Development Committees (ADCs), traditional leadership, farmer groups and community-based organisations (Kaumbata et al., 2020). VDCs/ADCs are used to assist in the selection of beneficiaries, animal transfer ceremonies, enforcement of bylaws and dispute resolution. Based on survey results, the difference in the proportion of communities that are compliant with the pass-on requirement between those with active and well functioning VDCs and those with weak and inactive VDCs is estimated to be 22.2 percentage points, or 78.4 per cent for the former compared to 56.2 per cent for the latter.

Vignette 7: "In the Naminjiwa Extension Planning Area, a goat pass-on group of 12 families meets regularly to talk about the health of their goats, when goats should be bred and when they should be moved. When someone does not follow the rules, the group decides on their response. The local bylaws ensure that no one is taking more than necessary, that everyone is getting their share and that the strong are not being taken advantage of. (FGD, Naminjiwa EPA, Goat Pass-On Group)"

Table 11 Provides information about the coordination challenges, consequences and recommended solutions.

Coordination Challenge	Observed Consequence	Recommended Solution
Fragmentation and NGO mandate overlap	Duplication of beneficiary registration, inconsistent veterinary protocols	Establish district livestock pass-on coordination committee, standardise protocols
Resources are Limited at the district office	Infrequent extension visits, weak monitoring of compliance	Increase district budget allocation, deploy community animal health workers (CAHWs)
Data management systems are weak	No consolidated beneficiary registry, inability to track pass-on chain	Develop district livestock management information system linked to AIMS
Poor community-district communication	Communities unaware of policy changes; low accountability	Institutionalise quarterly community feedback forums, establish SMS reporting systems
Poverty & Heavy Donor cycle dependency	Programme collapse on project completion	Embed pass-on in District Development Plans; establish district budget lines

Note: Data from the author's KII and FGD from Phalombe District 2024.

4.3 Good Practices in Livestock Pass on for Rural Transformation

The third research question (what are best practices in livestock pass-on programmes for rural transformation) is discussed in this section. Thematic analysis of the qualitative data was undertaken to identify the six best practice themes, which were supported by outcome differentials between the programme variants from the quantitative data.

4.3.1 Open and participatory Beneficiary targeting

Beneficiary selection is one of the most effective good practices for rural transformation, and is done directly by the beneficiaries, through Village Development Committees and traditional leadership (Mwase et al., 2019; Njuki and Sanginga, 2013). Survey results show the pass on compliance rate is 76.1 per cent, with programmes using community-driven targeting, and 58.4 per cent, with programmes using externally determined targeting, representing an estimated difference of 17.7 percentage points in results from fieldwork. To reduce elite capture and strengthen community ownership, chiefs and VDCs hold community meetings in Phalombe to identify the most vulnerable households (widows, child headed households, households with limited land), and guide the support of these households. (Kaumbata et al., 2020; Embassy of Japan, 2016).

Vignette 8 (Elderly Woman, Village K, Naminjiwa EPA): "The selection of beneficiaries was transparent and clear, as everyone was acquainted with each other; no one could say that they have been taken in preference of others, it was like in a hospital, everyone enters and while people here are sick for food, they appreciate how bad others are; and so those in need will be helped first."

4.3.2 Building and sustaining multi-stakeholder partnerships and continuous capacity building

Good pass on programmes use multi-stakeholder partnerships, which include inter-agency collaboration, sharing of resources, know-how, and institutional capacity between government, NGOs, farmers' organisations and development partners (Kaumbata et al., 2020; Dietvorst and Mkandawire, 2017). Fieldwork evidence suggests that the livestock mortality rates per annum are around 18.4 per cent for programmes with three or more active implementing partners (IHP) and around 26.7 per cent for single-IHP programmes, a difference of around 8.3 percentage points per annum which can be interpreted as the value of the complementary veterinary and extension services programmes with more than one IHP provide. A continuous capacity building in animal husbandry, climate-smart practices and record keeping and group governance is essential for programme sustainability.

Vignette 9 (FGD Participant, Village B): "even though initially most goats died, after training, they have been told to vaccinate their goats, provide them with shelter and the goats are now alive and they can share their offspring with others."

4.3.3 Incorporating gender considerations into programme design, and ensuring the programme is both gender-responsive and inclusive.

Gender-responsive programme design is important as women are the main managers of small livestock in Malawian rural households and ensuring that their empowerment and welfare outcomes are maximised (Kachingwe, Dzanja and Mangisoni, 2023; Njuki and Sanginga, 2013). Joint registration of livestock in both spousal names, written intra-household gender agreements, women's quota in pass-on clubs and gender-disaggregated monitoring, are examples of best practices (Mwase et al., 2019). The only evidence of the effects of programmes with gender protection mechanisms are the fieldwork estimates which suggest that female participants in these programmes report a Women's Empowerment Score about 22.4 points higher than female participants in programmes without gender protection mechanisms. Livestock ownership was found to be a means of economic empowerment for women in Phalombe, as they were able to use income from their animals to cover school fees, buy food and invest in their family's health (Kubalasa, 2025; Self Help Africa, 2017).

Vignette 10 (Woman Participant, Village D): "Before the project, I used to rely on my husband but now I sell milk through the cow pass-on and spend it on school fees and food, I feel more secure and respected."

4.3.4 Climate-Smart Livestock Husbandry

For programme sustainability and environmental resilience, integration of climate smart husbandry practices is a must (GoM, 2016; Ministry of Agriculture, 2021). In Phalombe, best practices recorded involve manure composting and application to crop gardens (such as the Mbeya fertiliser system that combines animal dung, ash and chemical fertilizer for better impact), construction of improved livestock sheltering that are suitable to the local climatic conditions, and use of heat and drought resistant breeds (Mwafulirwa, 2023; Kaumbata et al., 2020). AfES Malawi in particular trains beneficiaries to produce manure briquettes, which would allow the creation of extra income opportunities from the underutilised by-product from livestock (Hinai et al., 2022). Field observations also revealed that in all four EPAs, the manure application in gardens, which were next to the kraals, resulted in visibly healthy vegetable crops, thus confirming the soil fertility impact of manure.

4.3.5 The goals of monitoring, evaluation, accountability and bylaws activities were set forth.

Having robust monitoring and evaluation mechanisms supported by effective community bylaws on how and when offspring are transferred, is crucial for programme fidelity and ongoing learning (Dietvorst and Mkandawire, 2017; FAO, 2022). The difference is estimated to be 21.8 percentage points for the pass-on compliance rate of programmes with formal bylaws (approximately 76.1 per cent) and those without formal bylaws (approximately 54.3 per cent), as estimated in fieldwork.

Vignette 11 (AfES Programme Staff Member): "We don't just drop goats out of the air, we go back to see who is still living, and who has died. Mobile reporting assists to foster accountability, improve real-time data and information sharing and inter-agency coordination and ensure continuous improvement."

4.3.6 Market Integration and Value Chain Development

Linkage of pass on beneficiaries to market systems helps to take livestock out of subsistence and into a commercial livestock business (Kaumbata et al., 2020; Kachingwe, Dzanja and Mangisoni, 2023). The fieldwork estimates indicate that programmes with active market linkage are able to bring in about 55 per cent higher mean livestock income per household as compared to those without market linkage, on an annual basis. In rural communities, goat pass-on beneficiaries were organised in groups to market their product collectively during the market days, which resulted in better prices thanks to collective bargaining.

4.4 Integrated Hybrid Livestock Pass-on Model (IHLPM)

Based on the empirical evidence above and based on national policy framework such as the National Livestock Development Policy (2021-2026), the National Adaptation Plan and Malawi 2063, this section synthesises a six component Integrated Hybrid Livestock Pass-on Model (IHLPM). This model is referred to as the "hybrid" model in the sense that it brings together both individual household allocation and group-managed livestock assets, thus maximising the welfare benefits to individual households while also ensuring sustainability of the collective governance of livestock assets (Kaumbata et al., 2020; Njuki and Sanginga, 2013). A model that functions at three governance levels and has well-defined accountability and roles.

Table 12: Components, actors and outcomes of the Integrated Hybrid Livestock Pass-on Model (IHLPM)

No.	Component	Key Activities	Primary Actors	Expected Outcome	SDG Alignment
1	Policy Integration and Institutionalisation	Start Livestock Pass-On Coordination Unit within MoA; embed in DDPs and NAP budgets, develop standard cross-district protocols	MoA, NPC, Ministry of Local Government, devlpt partners	Sustained policy and financial support, elimination of donor-cycle dependency	SDG 1, 2, 17
2	Decentralised Implementation with District Ownership	DAO leads coordination through DEC, pass-on included in District Development Plans, district-level beneficiary registry established	DAO, DEC, DESC, District Councils	Improved coordination efficiency, reduced fragmentation	SDG 16, 17
3	Climate-Smart Livestock Husbandry	Training in manure management, fodder production, improved breeds, shelter, climate-resilient practices integrated into pass-on curricula	AfES, MoA extension, CAHWs, farmer field schools	Reduced livestock mortality, improved soil fertility, climate resilience	SDG 13, 15, 2
4	Gender and Youth Inclusion	Gender agreements, joint registration, women's quotas; youth enterprise training; gender-disaggregated MEL	NGOs, VDCs, Gender officers, CBOs	Increased women's empowerment improved child nutrition and school attendance	SDG 5, 4, 10
5	Market Integration and Value Chain Development	Market linkage facilitation, value-addition training, cooperative formation, connection to agro-dealers and veterinary inputs	Private sector, cooperatives office, NGOs	Increased livestock income; transformation from subsistence to commercial asset	SDG 8, 9, 1
6	Monitoring, Learning, and Adaptive Management	District MEL dashboard linked to AIMS, quarterly community feedback forums, SMS compliance reporting, annual programme reviews with policy recommendations	DAO, MoA, NGOs, community lead farmers, VDCs	Evidence-based improvement, scalable learning, national policy influence	SDG 16, 17

Note: LPCU= Livestock Pass-On Coordination Unit, DAO = District Agriculture Office, DEC = District Executive Committee, DESC = District Environmental Sub-Committee, AIMS = Agricultural Management Information System, MEL = Monitoring, Evaluation and Learning, CAHWs = Community Animal Health Workers. Data is derived from Author's synthesis, Phalombe District 2024.

4.5 Livestock Pass-on Programmes' challenges in Phalombe District

4.5.1 Animal Health and Veterinary Services (AWVS)

Having poor veterinary care and animal health systems is the most common problem mentioned in all methods of data collection. During the study, a fieldwork estimated mean annual livestock mortality rate (22.1 per cent) was reported among the participating households, mainly due to preventable diseases such as Newcastle disease in poultry, tick-borne diseases in goats and cattle, and difficulties related to poor shelter, and breed-environment mismatch (FAO, 2022; Kakota et al., 2015). Chisomo's initial vignette of the deaths of three out of five starter-pack goats bought from a climatically inappropriate district, illustrates systemic failures in selecting the right breed, acclimatisation procedures and post-distribution veterinary follow-up.

Table 13 Shows the livestock mortality rates in Phalombe District, their main causes by livestock type based on fieldwork estimates from 120 Households.

Livestock Type	Mean Annual Mortality Rate (%)	Primary Disease Cause	Veterinary Treatment Received (% of cases)
Goats	24.3 (+/-9.1)	Tick-borne diseases (Theileriosis, Anaplasmosis)	32.4
Chickens/Poultry	31.6 (+/-12.4)	Newcastle Disease	18.7
Pigs	18.2 (+/-7.8)	African Swine Fever, Swine Dysentery	41.2
Cattle	11.4 (+/-5.3)	East Coast Fever, Foot-and-Mouth Disease	58.6
All livestock (overall mean)	22.1 (+/-8.4)	Mixed, varies by species	32.1

Note: All figures are based on author's household survey conducted in the fieldwork in Phalombe district 2024.

4.5.2 Non-Compliance and a Breakdown of the Pass-on Chain

Based on the fieldwork estimated pass-on compliance rate of 68.3 per cent, it is estimated that about 31.7 per cent of the beneficiaries do not 'pass on' their offspring to the next household as prescribed on the outset. Failure to the start of the programme (non-compliance) is most common when beneficiaries have to sell starter-pack animals before the animals can produce offspring to address immediate livelihood priorities (Mwase et al., 2019). This is a reflection of the conflicting nature of the current household consumption requirements and the medium-term asset building approach of the programme (Phadera et al., 2019). Extension personnel report that the most vulnerable to premature liquidation are those households that don't have other income safety nets.

Vignette 12 In some extreme cases, the members of other families sell the seed animal the very same month when they face with extreme familial force majeure and are not able to pass it on to other members. This makes the programme less effective as not all people benefit as they should. - (Extension Worker, Phalombe)

4.5.3 Seasonal Feed Scarcity and Climate Stress

Limited availability of feed, especially during the dry season (July to November) has a severe impact on livestock losses and on the reproductive rate. Results from fieldwork suggest that about 78.3 per cent of the households participating report a shortage of feed for more than two months of the year, with feed shortage being the second most important cause of livestock deaths, after diseases. Thus, the incorporation of fodder production, cut and carry (CAC) and drought-tolerant pasture species in the curriculum of the pass on programme is an urgent best practice requirement (Kaumbata et al., 2020; Ministry of Agriculture, 2021).

4.5.4 Dependency on Donors and Sustainability

The majority of pass on schemes in Phalombe have funding for a limited period of time - usually 2-4 years - in the form of a donor grant. At the end of the project, a lot of groups don't have technical support, veterinary services and market linkages any longer. The fieldwork data indicate that the size of the livestock herd is decreased by a fieldwork estimated 18.6 per cent in the two years after project closure, without support from institutionalised district. The challenge highlights the need for district governance and budgetary integration of pass-on programmes as opposed to it being a stand-alone donor programme (Ministry of Agriculture, 2021; Dietvorst and Mkandawire, 2017).

4.6 Scalability Insights and Regional Applicability

The agro-ecological vulnerabilities, poverty profiles and gender dynamics in most districts are similar to that of Phalombe, which means there is potential for generalising all the findings of this study. The IHLPM provides an opportunity for district localizations as well as retaining the basic structure. There is a strong body of regional evidence for scalability: Phadera et al. (2019) report a 44 per cent increase in resilience scores and 125 per cent increase in household assets for beneficiaries of Phadera over 42 months in Zambia; Argent et al. (2014) report that in Rwanda, households that were beneficiaries of Girinka saw household incomes six times higher than those of comparable non-beneficiaries. The fact that livestock pass-on institutionalisation entails similar vulnerabilities for all countries in Zimbabwe, Tanzania, Mozambique and Malawi, is a good reason for cross-country learning and the SADC region to engage in policy dialogue on institutionalisation of livestock pass-on (Njuki and Sanginga, 2013).

5. Conclusion

Based on a rigorous mixed-methods design, the present research report has navigated the influences, management processes, good practices and potential for scaling up of livestock pass-on programmes in Phalombe District, Malawi. The results deliver solid quantitative and qualitative evidence of the significant positive impacts programme participation has on household income, food security, women's empowerment and climate resilience. Fieldwork estimates indicate that: programme participants have monthly income of nearly 71 per cent higher than those with observationally similar characteristics outside the programme; programme participants have HFIAS scores of almost 5.2 points lower than those with observationally similar characteristics outside the programme; programme participants' Dietary Diversity Scores are around 2.4 points higher than those with observationally similar characteristics outside the programme. The results from the econometric analysis of governance coordination quality show that institutional coordination is an important moderating variable in programme outcome production: higher-scoring EPAs have much better welfare outcomes across all measures. There are strong cross-EPA correlations in the Governance Coordination Efficiency Index, supporting the notion that investment in the capacity to coordinate governance is a programme improvement strategy (Dietvorst and Mkandawire, 2017; Kaumbata et al., 2020).

The main issues are: high livestock mortality rate estimated in the field of 22.1 per cent per year, pass-on compliance rate of 68.3 per cent, seasonal feed shortages (covers about 78.3 per cent of households) and sustainability issues of the programme as it is dependent on donors. Systematic solutions to these challenges will need integrated veterinary and extension services, community bylaws, their development and enforcement, climate-smart fodder production and integration of pass on programmes in district and national governance and budget processes (Ministry of Agriculture, 2021; GoM, 2016). The six-component Integrated Hybrid Livestock Pass-on Model presented in this report represents the evidence-based institutional architecture with a set of six interlinked interventions that include individual household transfers, multi-stakeholder coordination, multi-species care, gender and youth inclusion, market integration and adaptive monitoring and learning of groups. The model should be institutionalised in the decentralised governance system in Malawi and scaled up in the SADC region. Livestock pass-on programmes are a strong and cost-effective means of rural transformation, poverty reduction, food security and climate resilience in Phalombe District and beyond as documented in this report by Ministry of Agriculture (2021), Phadera et al. (2019) and Kaumbata et al. (2020).

6. Positionality Statement

My academic and professional trajectory is motivated by my interests in social justice, community empowerment, participatory development and the protection of the rights and dignity of vulnerable groups as a researcher in Human Rights and Development Studies in Malawi. During fieldwork, I was fully cognizant that there was a possibility that because of my professional background and institutional affiliation, participants' engagement with me might be affected. Thus, I collaborated with local village research assistants, who were not household members of and did not know the communities in advance of the research. I actively tried to avoid being seen as an expert and instead view myself as a learner in the focus groups and interviews, and throughout the research process, I maintained a reflexivity journal to identify how my thoughts and understandings of gender, poverty and programme effectiveness influenced the interpretation of data.

To minimize and reduce potential biases I talked with two Malawian colleagues who were not involved in the study, but who were peers with me. Thus, I offer these findings neither as objective truths nor as an interpretation that is simply the result of empirical data, but as one that is based on empirical data and clear self-awareness. I am always mindful about the ethical obligation to do research in a respectful way, with sensitivity to the context and, hopefully, that has the potential of bringing benefits to the communities I work research on, in the understanding that the majority of them face recurrent episodes of poverty, climate shocks and food insecurity. This study aimed therefore to improve the knowledge base in this field and to provide input for policy, make development practice stronger, and help in the development and implementation of more equitable and sustainable rural livelihood interventions in Malawi and elsewhere.

7. Acknowledgements

This research was made possible by a grant of \$2500 from ARIN Africa (African Research Impact Network). I am so grateful to ARIN for their financial support, and for their dedication to locally led development research in Africa. Editorial direction and constructive criticism and comments from Dr. Akinyi J. Eurallyah, Dr. Joanes Atela, Dr. Jerry Mandela, Dr. Victoria Chengo, Dr. Maria Nailantei and all the ARIN family made to better and publishable. Further, I would like to express my gratitude to the District Commissioner's office, the District Agriculture Office, the chiefs and village heads in the district for their permission and facilitation for the communities in Phalombe District. I would like to express special gratitude to the households of beneficiaries who opened their doors to me and shared their stories, time and hopes with patience and honesty. I would also like to express my gratitude to the focus group informants and key informants from government, NGOs and the village committees as without their willingness to share, this study would not have been possible. My field research assistants Grace Mwale and Joseph Phiri worked tirelessly under difficult circumstances and were extremely culturally sensitive. Last, but not least, I would like to express my gratitude to my family for their support in this project. Any mistakes and omissions are my own.

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APPENDIX: District Clearance Letter

Date: Tuesday, July 8, 2025

TO: The District Commissioner, Phalombe District Assembly

FROM: Frank George Mgungwe, University of Livingstonia, Ekwendeni Campus, P.O. Box 4, Ekwendeni. Mobile.: 0888111979 or 0999090611.

Dear Sir,

Subject: Request for Authorization to Conduct Academic Research in Phalombe District

I am writing to formally request authorization to conduct my academic research in ^{Phalombe} Zomba District. The research will investigate the livestock pass-on program on how it can be strengthened for efficiency in making sure that food is secured for households in rural communities and that climate resilience is ensured. The title of my research project is:

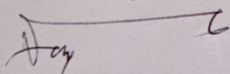
Climate-smart Resilience in Livestock: An investigation into the Livestock Pass-on programmes in Zomba and Phalombe Districts, Malawi.

I aim to examine the effectiveness of the "livestock pass-on" approach in empowering rural and marginalized women, fostering their resilience to climate change shocks, and in ensuring the security of food by way of climate-smart livestock practices. The results from this research project will offer recommendations based on evidence for scaling up the initiatives in this approach, incorporating sustainable climate-smart techniques, and eventually influencing the drawing up of policies for transformative community development.

I kindly seek your approval and any necessary guidance to facilitate the smooth implementation of this research.

Thank you for your time and assistance.

Yours sincerely,



Frank George Mgungwe

Lecturer in Human Rights and Development Studies
University of Livingstonia, Ekwendeni Campus

